Kaushik JV Design System v2

Zillennial Bridge | Heritage Cream x Cerulean Teal

A design engineering system that bridges Tactile Nostalgia (1995-2005) with Apple's 2026 Liquid Glass minimalist standards. Built for a portfolio that signals dual fluency in design craft and front-end engineering.

VERSION 2.0

TECH STACK

Framework	Next.js 14 (App Router) + TypeScript
Styling	CSS Custom Properties + Tailwind CSS
Animation	Framer Motion -- Spring physics
Fonts	Fraunces Playfair Display DM Serif Display Space Mono
Tokens	W3C Design Tokens format (tokens.json)
Repository	github.com/JVK0804/jvk-design-system-figma-to-code-by-claude

CONTENTS

01 -- Design Philosophy

The design system operates on three core principles that govern every decision -- from colour selection to animation timing. Each principle is load-bearing; removing any one collapses the aesthetic contract.

1

Tactile Skeuomorphism 2.0

Modern glassmorphism mimics the weight and texture of physical hardware -- translucent plastics, CRT-style glows, grain on every surface. Physical weight is achieved through layered translucency, not literal texture mapping.

2

Intentional Friction

Interactions have a physical press phase (30ms compress to scale(0.955)) and a spring-return (280ms overshoot to scale(1.018)). Mechanical feel, fluid 120Hz execution. The "click-clack" is a design decision, not an animation afterthought.

3

The Hybrid Palette

Heritage Cream base (#E8E4D9) evoking 90s computer beige, paired with High-Voltage Cerulean Teal (#0E7490) evoking fiber-optic infrastructure. Two eras on one surface.

02 -- Colour Tokens

All backgrounds carry a warm brown undertone. Never substitute with neutral or cool greys -- this is what makes the system feel human rather than clinical. The accent (#0E7490) appears on one semantic layer only: interactive elements.

BASE -- HERITAGE CREAM

	--color-heritage-cream	#E8E4D9	*Page base, scene background*
	--color-cream-bright	#F5F2EB	*Card highlight, specular layer*
	--color-cream-shadow	#D4CFBF	*Depth, dividers, borders*

ACCENT -- CERULEAN TEAL

	--color-primary	#0E7490	*Primary interactive, CTAs*
	--color-primary-hover	#22B8CF	*Hover states, CRT glow*

--color-primary-phosphor	#A5F3FC	*CRT terminal text on dark*
--color-primary-subtle	rgba(14,116,144,0.10)	*Tag fills, badge backgrounds*
--color-primary-border	rgba(14,116,144,0.28)	*Tag and badge borders*

INK -- TEXT RAMP

--color-ink	#1C1A14	*Primary text on cream*
--color-ink-muted	#5A5648	*Body copy, descriptions*
--color-ink-disabled	#9A9488	*Captions, metadata, read-time*

CONTRAST RATIOS (ON #E8E4D9)

Colour	Ratio	WCAG	Role
#0E7490 on cream	4.8:1	AA	Body text safe
#1C1A14 on cream	12.1:1	AAA	Primary text
#5A5648 on cream	5.2:1	AA	Body copy safe

03 -- Typography Scale

Display and H1 use weight 200 (Extra Light). This reads as open and approachable rather than cold. Weight 500 is reserved for structural work: H3, labels, and CTAs.

FONT FAMILIES

Token	Family	Role	Source
--font-display	SF Pro Display	Hero/H1 sans	System
--font-body	SF Pro Text	H2 to body sans	System
--font-mono	SF Mono	Terminal labels	System
--font-mono-space	Space Mono	CRT labels, mono	Google Fonts
--font-serif-fraunces	Fraunces	Retro-futurist display	Google Fonts
--font-serif-playfair	Playfair Display	Editorial display	Google Fonts
--font-serif-dm	DM Serif Display	Tech-editorial display	Google Fonts

TYPE SCALE

Name	Size	Weight	Tracking	Line-H	Usage
Display	80px	200	-0.05em	1.0	Hero headline
H1	52px	200	-0.04em	1.1	Section hero

H2	28px	400	-0.03em	1.2	Section titles
H3	20px	500	-0.02em	1.3	Card titles
Body	16px	400	-0.01em	1.75	Body copy
Caption	13px	400	0em	1.5	Metadata, read-time
Label	11px	500	+0.08em	1.4	UI labels (CAPS)
Terminal	13px	400	+0.12em	1.6	CRT terminal text (CAPS)

04 -- Spacing System

Base unit: 4px. All spacing values are multiples of 4. Never use arbitrary values outside this scale.

Token	Value	Semantic Role
--space-1	4px	Icon gaps, micro spacing
--space-2	8px	Inline elements
--space-3	12px	Tight stacks
--space-4	16px	Component padding
--space-6	24px	Card internal spacing
--space-8	32px	Between components
--space-12	48px	Section gap (mobile)
--space-20	80px	Section gap (desktop)
--space-28	112px	Hero breathing room

05 -- Border Radius

Concentric Apple logic: outer container radius is always larger than the inner element radius. This creates a visually consistent layered depth.

Token	Value	Usage	Component
--radius-sm	4px	Tags, chips, badges	Tag component
--radius-md	8px	Buttons, inputs	HapticButton
--radius-lg	14px	Project cards	GlassCard
--radius-xl	20px	Modals, scene frames	Scene container
--radius-pill	100px	Label pills, CTA buttons	Pill variant

06 -- Glass Card Component

The project card is a five-layer composition. Order matters precisely. The grain lives on the scene, not the card -- this makes the grain read as the world's texture, so the card appears to float on a physical surface rather than being printed into it.

LAYER ANATOMY

#	Layer	Description
1	Scene Grain	SVG feTurbulence on scene -- zero HTTP requests, ~0.4kb inline
2	Heritage Cream	#E8E4D9 base scene background -- warm brown undertone always present
3	Glass Card	rgba(232,228,217,0.45) + backdrop-filter: blur(12px)
4	Specular ::before	linear-gradient(135deg, white/28%, white/4%) -- light from top-left
5	Glint ::after	1px top edge: transparent -> white/80% -> transparent
6	Content	z-index: 20, relative positioning -- all children go here

GLASSCARD.TSX -- USAGE

```
  import { GlassCard } from './components/GlassCard'

// Basic
<GlassCard>
  <h3>Project Title</h3>
  <p>Description</p>
</GlassCard>

// With grain owned by card (if no parent scene)
<GlassCard withGrain>
  <h3>Project Title</h3>
</GlassCard>
```

CSS CUSTOM PROPERTIES USED

```
  --color-glass-bg:            rgba(232, 228, 217, 0.45)
--color-glass-specular-from: rgba(255, 255, 255, 0.28)
--color-glass-specular-to:   rgba(255, 255, 255, 0.04)
--color-glass-glint:         rgba(255, 255, 255, 0.80)
--border-glass:              1px solid rgba(255, 255, 255, 0.55)
--blur-glass:                blur(12px)
```

07 -- Haptic Spring -- Click-Clack

The mechanical button interaction mimics the physical sensation of pressing a tactile switch. Two phases, each with a distinct timing and easing curve. Always bind to onPointerDown, not onClick -- the 60-80ms difference is the entire tactile illusion.

TIMING BREAKDOWN

Phase	Duration	Transform	Easing
Press (compress)	30ms	scale(0.955)	cubic-bezier(0.4, 0, 1, 1) -- hard ease-in
Hold	20ms	static	--
Spring return	280ms	scale(1.018->1.0)	Spring: stiffness 520, damping 22, mass 0.6
Settle	50ms	scale(1.0)	micro-damp

USEHAPTICPRESS.TS -- HOOK IMPLEMENTATION

```
 const springConfig = {
  type: 'spring',
  stiffness: 520,  // high -- gives snap-back authority
  damping:   22,   // low  -- allows 1.018 overshoot (the "clack")
  mass:       0.6  // keeps total duration under 320ms for 120Hz
}

export function useHapticPress() {
  const controls = useAnimation()

  const onPointerDown = async () => {
    await controls.start({
      scale: 0.955,
      transition: { duration: 0.03, ease: [0.4, 0, 1, 1] }
    })
  }
  const onPointerUp = async () => {
    await controls.start({ scale: 1.018, transition: springConfig })
    await controls.start({ scale: 1, transition: { duration: 0.05 } })
  }

  return { controls, onPointerDown, onPointerUp }
}
```

08 -- Button Variants

All six button states share the same HapticButton component, controlled via the variant prop. The haptic spring is disabled when variant="disabled".

Variant	Background	Text	Border	Notes
primary	#0E7490	#F5F2EB	none	Default CTA -- radius-md (8px)
pill	#0E7490	#F5F2EB	none	Full pill -- radius-pill (100px)

hover	#22B8CF	#F5F2EB	none	Active/hover state of primary
secondary	transparent	#0E7490	#0E7490	Outlined -- 1px solid teal
ghost	transparent	#0E7490	none	No border, subtle hover tint
disabled	#D4CFBF	#9A9488	none	cursor: not-allowed, opacity 0.7

USAGE

```
  import { HapticButton } from './components/HapticButton'

<HapticButton variant="primary">View Project</HapticButton>
<HapticButton variant="pill">Get in Touch</HapticButton>
<HapticButton variant="secondary">Learn More</HapticButton>
<HapticButton variant="ghost">See All Work</HapticButton>
<HapticButton variant="disabled">Coming Soon</HapticButton>
```

09 -- Tags & Badges

Tags (size="md") and Badges (size="sm") share the same Tag component. Both use --color-primary-subtle fill (teal 10%) and --color-primary-border stroke (teal 28%). Badges add uppercase + letter-spacing for a stamp-like label feel.

Size	Height	Font	Padding	Notes
md	28px	13px	0 12px	Normal case -- no forced letter-spacing
sm	22px	11px	0 10px	Uppercase -- +1.5px letter-spacing (badge style)

USAGE

```
  import { Tag } from './components/Tag'

<Tag size="md">Design Systems</Tag>
<Tag size="md">Zillennial</Tag>
<Tag size="sm">NEW</Tag>
<Tag size="sm">LOCKED</Tag>
<Tag size="sm">Phase 1</Tag>
```

10 -- File Structure & Usage

DIRECTORY TREE

```
  figma design system code/
|-- styles/
|   `-- tokens.css          <- CSS custom properties (source of truth)
|-- tokens/
|   `-- tokens.json         <- W3C Design Tokens (Figma + Tailwind)
|-- hooks/
|   `-- useHapticPress.ts   <- Framer Motion spring hook
|-- components/
|   |-- GlassCard.tsx       <- Five-layer glass card primitive
|   |-- GrainOverlay.tsx    <- Performance-safe SVG grain layer
|   |-- HapticButton.tsx    <- Button with 6 states via variant prop
|   `-- Tag.tsx             <- Tag (md 28px) and Badge (sm 22px)
`-- tailwind.config.ts      <- Tailwind extended with all design tokens
```

IMPLEMENTATION ORDER

1. Drop tokens.css into /styles/ and import in app/globals.css
2. Paste tailwind.config.ts theme.extend into your tailwind.config.js
3. Build GrainOverlay -- it wraps everything else in the scene
4. Build GlassCard with grain wired in via withGrain prop
5. Build useHapticPress hook
6. Build HapticButton wrapper using the hook
7. Build the Project Card using GlassCard + HapticButton
8. Build the navigation shell with --blur-nav vibrancy

11 -- System Status

Layer	Status
Colour tokens	Locked
Typography scale -- SF Pro	Locked
Typography scale -- Serif v1.1	Locked
Spacing grid	Locked
Border radius	Locked
Glass card anatomy	Locked
Haptic spring values	Locked
tokens.css	v2 complete
tokens.json	v2 complete
GlassCard.tsx	v2 complete
HapticButton.tsx	v2 complete
useHapticPress.ts	v2 complete

GrainOverlay.tsx	v2 complete
Tag.tsx	v2 complete
tailwind.config.ts	v2 complete
Navigation shell	Phase 2
Full component library	Phase 2

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